Code No: 117CF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) "Design patterns solve many of the day-to-day problems faced by the Object-Oriented-designers and in many different ways". Explain how?
b) How the design pattern solves the challenges lying in applying the reusable software? Explain. [8+7]

2.a) What is Design pattern? Describe the step by step approach to apply a design pattern effectively.
b) In what ways design patterns and frameworks share similarities and in which ways they differ? [7+8]

3. Explain the following design problems in detail.
a) Embellishing the User Interface.
b) Spelling Checking and Hyphenation. [7+8]

4.a) What are creational patterns? Write in detail about Singleton and Prototype patterns.
b) Explain in detail about issues when applying the Factory Method pattern. [8+7]

5.a) What are the applications of Flyweight Pattern? Explain the structure and participants of Flyweight pattern with suitable example.
b) Differentiate between adapter and bridge design pattern. [8+7]

6.a) What is the intent and motivation of Proxy pattern? Explain.
b) Explain about the structure and participants of Decorator pattern. [7+8]

7.a) What are the implementation issues to be considered about Chain of Responsibility behavioral patterns? Explain.
b) Discuss about the structure and participants of Interpreter pattern. [8+7]

8.a) Describe about Intent, applicability and sample code of strategy pattern.
b) Explain about visitor pattern in detail. [8+7]

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